



Faculty of Computing
University of Sri Jayewardenepura

CCS4082 | CSE4092 – Bioinformatics

Phylogenetic Analysis of Dengue Virus Genomes Using the Galaxy Platform

Group Members

Student Name	Index No
H.K.S. Peiris	FC212045
J.M.D.M. Sulakshan	FC212026
E.W.U. Kalyana	FC211014
W.A.Y.L. Wickramasinghe	FC212004
H.E.M.U Srimal	FC212025

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1. Introduction

1.1 Background and Clinical Significance of Dengue Virus

The Dengue virus (DENV) is an important pathogen among the arthropod-borne viruses that cause infection in human beings living in the tropical and subtropical parts of the world. The virus is a single-stranded, positive-sense RNA virus, and it belongs to the family Flaviviridae and the genus OrthoFlavivirus. Transmission of the virus to human beings occurs mainly through the bite of an infected *Aedes aegypti* mosquito, while *Aedes albopictus* serves as the secondary vector. DENV infection can lead to either a mild, self-limiting febrile illness referred to as dengue fever or severe forms of the disease like dengue hemorrhagic fever and dengue shock syndrome.

There are four antigenic types of DENV, including DENV-1, DENV-2, DENV-3, and DENV-4. Infection by one type creates lifelong immunity against that serotype but not the other three. However, infection by another serotype increases the chances of developing severe disease through a process known as antibody-dependent enhancement. Thus, the diversity of serotypes coupled with the high mutation rate of the virus makes the molecular characterization of DENV circulating strains very important.

1.2 Role of Phylogenetics in Viral Epidemiology

Phylogenetic studies play a vital role in molecular epidemiology and monitoring outbreaks. This is because RNA viruses reproduce using a high-fidelity RNA-dependent RNA polymerase, causing them to mutate at very high rates, thereby creating genetic variation which can be studied to reveal the history of the evolution of these viruses. From comparison of genome sequences of isolates from different geographic and temporal locations, it will be possible to understand the transmission routes of these viruses, recognize the introduction of new genotypes into a particular area, and follow up on the evolution of the virus over time.

1.3 Computational Workflow for Evolutionary Reconstruction

A typical bioinformatics pipeline for virus phylogenomic consists of four main steps. Firstly, genomic sequences of viruses are downloaded from a publicly available database like the NCBI Virus database. Secondly, these sequences need to be aligned to allow the comparison of homologous nucleotides among all sequences. In this research, MAFFT (Multiple Alignment using Fast Fourier Transform) was chosen to carry out the alignment process since it is highly suitable for long sequences of nucleotides. Thirdly, the structure of trees is inferred through maximum likelihood estimation and ModelFinder and ultrafast bootstrapping (UFBoot). These techniques are implemented by the program IQ-TREE. Finally, the whole pipeline has been conducted on the Galaxy platform.

1.4 Study Objectives

The aim of this research was to build and analyze a phylogenetic tree using 28 full-length genomes of DENV originating from human blood in Asia.

In particular, the investigation included the following objectives:

- To prove monophyly of DENV serotypes in the studied sample
- To reveal geographic and temporal clustering of the Asian strains
- To test robustness of tree branches with 100 vs. 1,000 bootstraps (extra credit)
- To explain evolutionary and epidemiological significance of clustering.

2. Materials and Methods

2.1 Sequence Retrieval and Dataset Selection

The complete genome sequences of DENV were obtained from NCBI Virus database (<https://www.ncbi.nlm.nih.gov/labs/virus/>) with the application of the following filter criteria: Taxonomy: Dengue virus (taxid:11053), Sequence Type: Complete genome, Host: Homo sapiens, Isolation Source: blood, and Geographic Region: Asia. The twenty-eight representative genomes of all four serotypes have been chosen for further analysis and are representative of 13 Asian countries including Singapore, Vietnam, China, India, Thailand, Myanmar, Pakistan, Sri Lanka, Bangladesh, Malaysia, Indonesia, Cambodia, and Taiwan between 1999 and 2023 (Table 2).

The screenshot displays the NCBI Virus database search interface. At the top, the NIH logo and 'National Library of Medicine' are visible. Below this, the 'NCBI Virus' logo and 'Sequences for discovery' tagline are present. A navigation bar includes links for 'Find Data', 'Help', 'How to Participate', 'Submit Sequences', and 'Contact Us'. A section titled 'Explore Virus Data' lists various viruses, including Influenza virus, Dengue virus, Zika virus, and SARS-CoV-2 coronavirus. Below this, a 'Refine Results' section shows applied filters: Geographic Region (1), Virus/Taxonomy (1), Nucleotide Completeness (1), Host (1), and Tissue/Specimen/Source (1). The 'Selected Results' section shows 0 / 2,047 results. A table of results is displayed with columns: Accession, Organism Name, Species, Genotype, Isolate, Length, and Nuc Completeness. The first result is PX460918.1, dengue virus type 2, Orthoflavivirus denguei, 2, KMH2023, 10723, complete.

Accession	Organism Name	Species	Genotype	Isolate	Length	Nuc Completeness
PX460918.1	dengue virus type 2	Orthoflavivirus denguei	2	KMH2023	10723	complete

Figure 1A. NCBI Virus database search interface showing the applied filters (taxonomy, host, isolation source, and geographic region).

The screenshot shows the NCBI Virus database search interface with the following filters applied:

- Refine Results:** Expand All | Collapse All
- Identifiers and Classification (5):**
 - Virus/Taxonomy: Dengue virus, taxid:12637
 - Accession: +
 - GenBank/RefSeq: +
 - Genotype: +
 - Isolate/Strain Name: +
- Location and Source (5):**
 - Nucleotide Completeness: complete
 - Assembly Completeness: +
 - Ambiguous Characters: +
 - Geographic Region: Asia
 - Host: Homo sapiens (human), taxid:9606
 - Tissue/Specimen/Source: blood

Figure 1B. NCBI Virus database search interface showing the applied filters (taxonomy, host, isolation source, and geographic region).

	Accession	Description	Genome_status	Region	Country	Collection_Date
1	FJ639691.1	Dengue virus isolate	Complete Genome	Asia	Cambodia	2007
2	KM403622.1	Dengue virus isolate	Complete Genome	Asia	Singapore	2013
3	PV083756.1	Dengue virus isolate	Complete Genome	Asia	China	2014-09
4	PV752120.1	Dengue virus isolate	Complete Genome	Asia	Sri Lanka	2017-02-03
5	MH110586.1	Dengue virus isolate	Complete Genome	Asia	China	2017-09
6	KY427084.1	Dengue virus isolate	Complete Genome	Asia	India	2010-09-01
7	KM217156.1	Dengue virus isolate	Complete Genome	Asia	Pakistan	2011-08
8	PQ465587.1	Dengue virus isolate	Complete Genome	Asia	Malaysia	2015-12
9	PQ657766.1	Dengue virus isolate	Complete Genome	Asia	Bangladesh	2023-09
10	EU687250.1	Dengue virus isolate	Complete Genome	Asia	Vietnam	2007
11	MG272274.1	Dengue virus isolate	Complete Genome	Asia	India	2016
12	KP792537.2	Dengue virus isolate	Complete Genome	Asia	Singapore	2011-06
13	KY672959.1	Dengue virus isolate	Complete Genome	Asia	China	2015-08-07
14	PX368925.1	Dengue virus isolate	Complete Genome	Asia	Indonesia	2019-12-25
15	PV344375.1	Dengue virus isolate	Complete Genome	Asia	Thailand	2015-07-28
16	PQ555702.1	Dengue virus isolate	Complete Genome	Asia	Malaysia	2023-09
17	KY670635.1	Dengue virus isolate	Complete Genome	Asia	Taiwan	2003-10
18	OP410995.1	Dengue virus isolate	Complete Genome	Asia	Singapore	2014-10
19	PX125311.1	Dengue virus isolate	Complete Genome	Asia	Indonesia	2022-12-20
20	ON115814.1	Dengue virus isolate	Complete Genome	Asia	Bangladesh	2021-08
21	GU363549.1	Dengue virus isolate	Complete Genome	Asia	China	2009-08-06
22	MN253125.1	Dengue virus isolate	Complete Genome	Asia	India	2017-10-27
23	FJ461326.1	Dengue virus isolate	Complete Genome	Asia	Vietnam	2007
24	KF955461.1	Dengue virus isolate	Complete Genome	Asia	Cambodia	1999
25	JF459993.1	Dengue virus isolate	Complete Genome	Asia	Myanmar	2002
26	KJ755855.1	Dengue virus isolate	Complete Genome	Asia	India	2013-07
27	FJ410280.1	Dengue virus isolate	Complete Genome	Asia	Vietnam	2008
28	PQ357540.1	Dengue virus isolate	Complete Genome	Asia	Thailand	2019-06-15

Figure 2: Metadata of the Downloaded 28 Genome Sequences

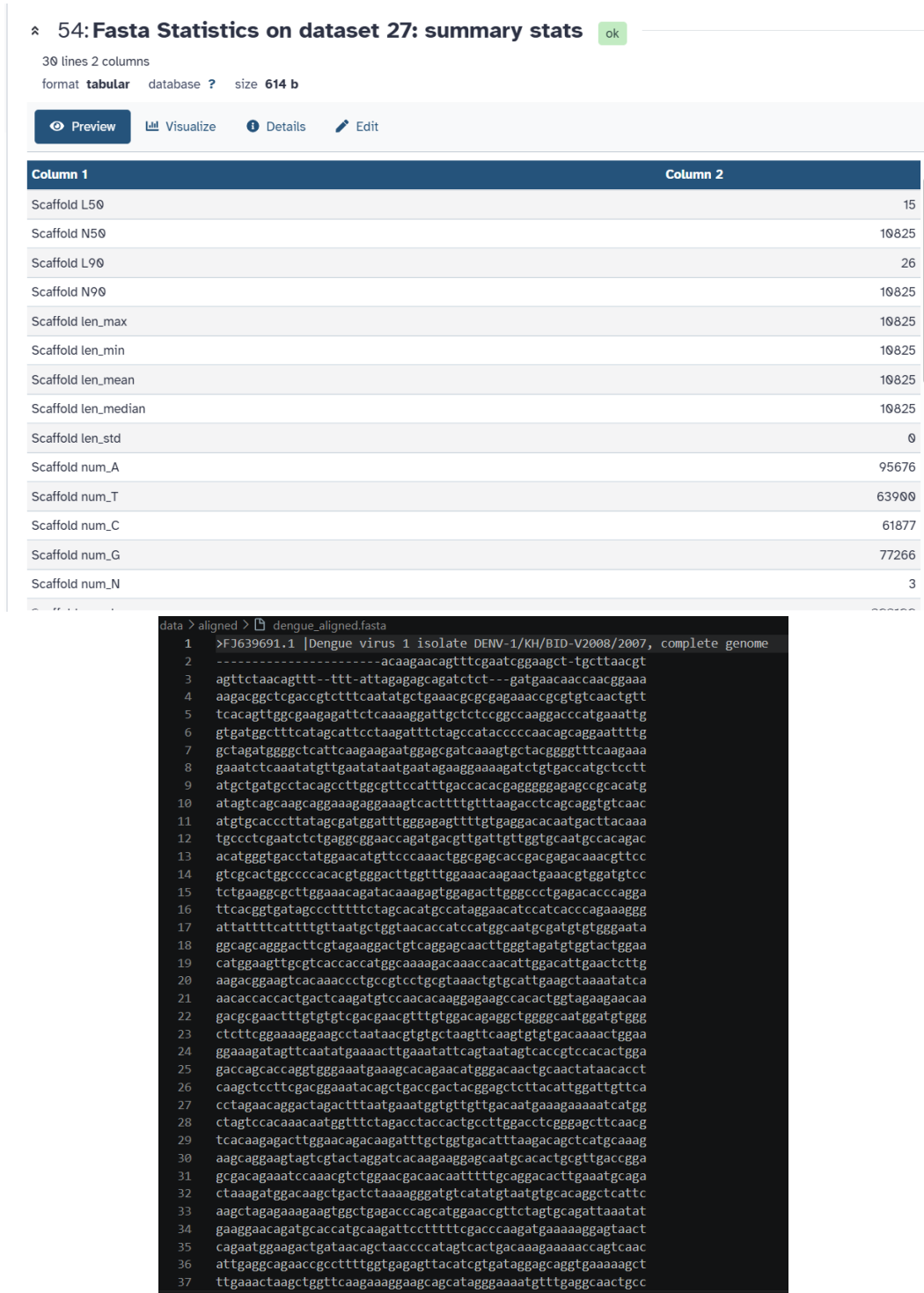


Figure 3: Output From Running FASTA Statistics Tool from Galaxy on the MAFIT Aligned FASTA file

2.2 Multiple Sequence Alignment

The 28 complete genomes were aligned using MAFFT through the Galaxy website, where default parameters were used, which means that the tool selects the best alignment algorithm depending on sequence count and length. The generated alignment was exported as FASTA file and evaluated for even length and gap presence before building a phylogenetic tree.

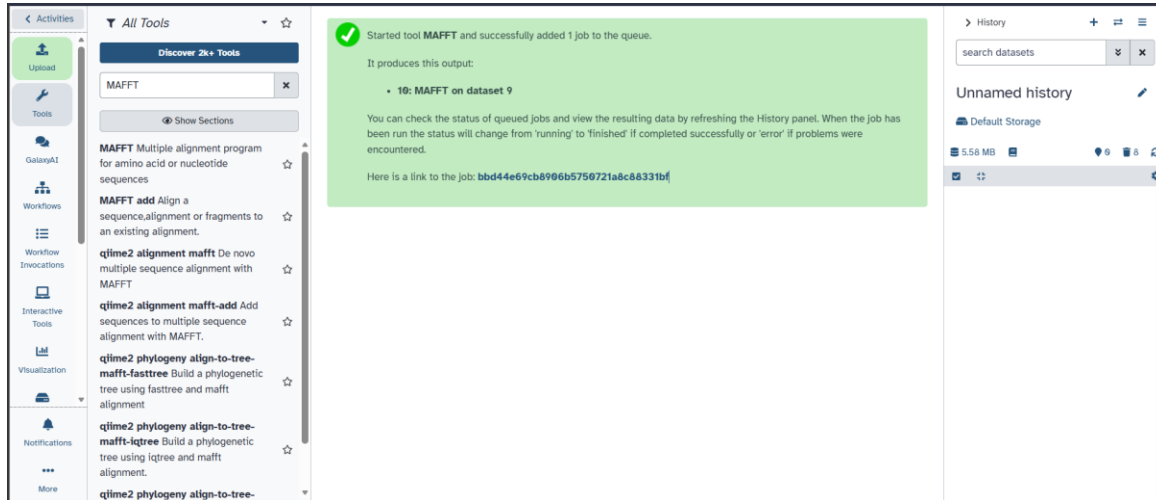


Figure 4. MAFFT alignment workflow execution within Galaxy.

2.3 Model Selection and Phylogenetic Reconstruction

Phylogenetic tree construction was done with the help of IQ-TREE 2.4.0 software through Galaxy using Maximum Likelihood approach. ModelFinder tool which pre-loaded in IQ-TREE software was initially used to determine the best nucleotide substitution model based on AIC, AICc, BIC criteria; TIM2+F+I+R3 was identified as the best model fit (Table 3). UFBoot2 was applied for measuring the reliability of each node in the tree. Two bootstrap analyses, one with 100 bootstrap replicates and another with 1,000 bootstrap replicates, have been done as per bonus requirement. The final output tree has been drawn without rooting, with isolate FJ639691.1 (DENV-1 strain from Cambodia) as reference.

Table 1. Parameters for Phylogenetic Reconstruction using IQ-TREE

Parameter	Setting
Sequence type	DNA
Tree method	Maximum Likelihood
Model selection	ModelFinder (AIC/AICc/BIC) followed by tree inference
Bootstrap method	Ultrafast bootstrap (UFBoot2)
Bootstrap replicates	1000

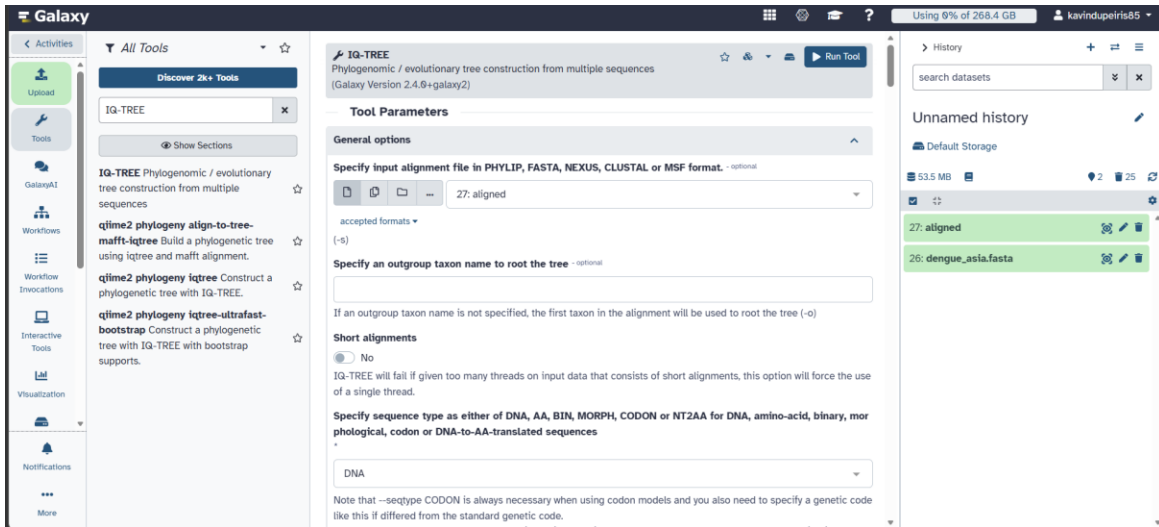


Figure 5. IQ-TREE / ModelFinder output showing selection of the TIM2+F+I+R3 substitution model.

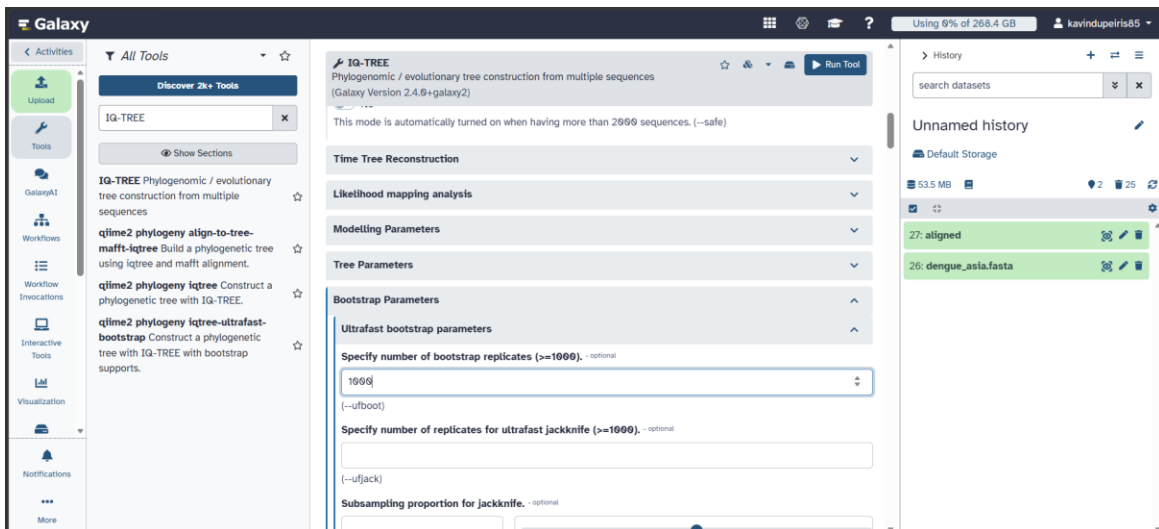


Figure 6. Bootstrap configuration used for the ultrafast bootstrap run.

3. Results

3.1 Dataset Characteristics and Geographic Distribution

Analysis was performed on 28 full human-origin DENV genome sequences from 13 different countries/territories from Asia, collected between the years 1999 to 2023. The dataset was equally distributed across all the four serotypes of the virus, containing 7 sequences for DENV-1, DENV-2, DENV-3, and DENV-4 respectively (Table 2). The average length of the sequence before alignment was around 10,650 bp.

Table 2. Summary of the 28 Dengue virus genome sequences used in this analysis.

Accession	Serotype	Country / Region	Collection Year
FJ639691.1	DENV-1	Cambodia	2007
KM403622.1	DENV-1	Singapore	2013
FJ410280.1	DENV-1	Vietnam	2008
PQ357540.1	DENV-1	Thailand	2019
JF459993.1	DENV-1	Myanmar	2002
PV083756.1	DENV-1	China (Guangdong)	2014
KJ755855.1	DENV-1	India	2013
PV752120.1	DENV-2	Sri Lanka (Colombo)	2017
PQ465587.1	DENV-2	Malaysia (Selangor)	2015
KY427084.1	DENV-2	India (Kerala)	2010
KM217156.1	DENV-2	Pakistan	2011
MH110586.1	DENV-2	China (Hangzhou)	2017
EU687250.1	DENV-2	Vietnam	2007
PQ657766.1	DENV-2	Bangladesh (Chattogram)	2023
OP410995.1	DENV-3	Singapore	2014
FJ461326.1	DENV-3	Vietnam	2007

Accession	Serotype	Country / Region	Collection Year
PX125311.1	DENV-3	Indonesia (Bali)	2022
KF955461.1	DENV-3	Cambodia	1999
ON115814.1	DENV-3	Bangladesh	2021
GU363549.1	DENV-3	China (Guangzhou)	2009
MN253125.1	DENV-3	India	2017
MG272274.1	DENV-4	India	2016
PX368925.1	DENV-4	Indonesia (Bali)	2019
KY670635.1	DENV-4	Taiwan	2003
PQ555702.1	DENV-4	Malaysia (Negeri Sembilan)	2023
KP792537.2	DENV-4	Singapore	2011
PV344375.1	DENV-4	Thailand	2015
KY672959.1	DENV-4	China (Yunnan)	2015

3.2 Multiple Sequence Alignment Statistics

The MAFFT alignment produced a final alignment length of 10,825 bp, reflecting insertions and deletions introduced during the alignment process. Of these sites, 46.82% were constant and 53.19% were variable, with the majority (50.77% of all sites) being parsimony informative. GC content was highly consistent across the 28 genomes, ranging from 45.63% (isolate PQ465587.1, DENV-2, Malaysia) to 47.67% (isolate MG272274.1, DENV-4, India), with a mean GC content of 46.60%. Gap content was minimal (approximately 1.3% per sequence on average), indicating a high-quality multiple sequence alignment and substantial sequence conservation despite the presence of distinct DENV serotypes.

Table 3. Multiple sequence alignment statistics.

Alignment Metric	Value	% of Sites
Total aligned length	10,825 bp	-
Constant sites	5,067	46.82%

Alignment Metric	Value	% of Sites
Variable (polymorphic) sites	5,758	53.19%
Parsimony-informative sites	5,496	50.77%
Singleton sites	262	2.42%
Mean GC content (range)	46.60% (45.63–47.67%)	-

53: Fasta Statistics on dataset 26: summary stats ok

30 lines 2 columns
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[Preview](#) [Visualize](#) [Details](#) [Edit](#)

Column 1	Column 2
Scaffold L50	14
Scaffold N50	10690
Scaffold L90	26
Scaffold N90	10638
Scaffold len_max	10735
Scaffold len_min	10176
Scaffold len_mean	10668
Scaffold len_median	10690
Scaffold len_std	100
Scaffold num_A	95676
Scaffold num_T	63900
Scaffold num_C	61877
Scaffold num_G	77266
Scaffold num_N	3

Figure 7A. FASTA statistics for the original (unaligned) sequence set.

54: Fasta Statistics on dataset 27: summary stats ok

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Column 1	Column 2
Scaffold L50	15
Scaffold N50	10825
Scaffold L90	26
Scaffold N90	10825
Scaffold len_max	10825
Scaffold len_min	10825
Scaffold len_mean	10825
Scaffold len_median	10825
Scaffold len_std	0
Scaffold num_A	95676
Scaffold num_T	63900
Scaffold num_C	61877
Scaffold num_G	77266
Scaffold num_N	3

Figure 7B. FASTA statistics for the MAFFT-aligned sequence set.

3.3 Best-Fit Substitution Model

According to the AIC and BIC standards, ModelFinder identified the TIM2+F+I+R3 as the best fit nucleotide substitution model with a value of 157,032.409 for AIC and 157,498.944 for BIC. The model showed an extremely high C<->T transition rate (29.4081), compared to other types of substitutions, reflecting transition bias in the genome evolution of DENV with a nucleotide composition that is predominantly adenine (32.03%). It was found that about 30.87% of the alignment positions were invariable, whereas the rest were divided into three Free Rate groups (Table 4).

Table 4. Substitution model parameters selected by ModelFinder (TIM2+F+I+R3).

Parameter	Value
Best-fit model (AIC / BIC)	TIM2+F+I+R3
AIC score	157,032.409
BIC score	157,498.944
Transition A↔G	9.3054
Transition C↔T	29.4081
Transversion rates (C↔G, G↔T)	1.0000 (reference)
Base frequencies (A / C / G / T)	32.03% / 20.71% / 25.87% / 21.39%
Proportion of invariable sites (I)	30.87%
Free Rate categories (3)	25.76% 0.147; 32.66% 1.674; 10.71% @ 3.877
Log-likelihood (lnL)	-78,448.5451 (SE 627.5467)
Total tree length	3.9970 substitutions/site

3.4 Maximum Likelihood Tree Reconstruction and Topology

Tree based on Maximum Likelihood had a log likelihood of -78,448.5451 (SE = 627.5467) with a total tree length of 3.9970 substitutions per site where 3.5830 (89.64%) were internal branches. All 28 sequences were divided into four monophyletic clades of serotypes (DENV-1 to DENV-4) each having 100% bootstrap support at their respective ancestral nodes. Two clades (DENV-2 & DENV-4) were paired together with 100% support where DENV-3 had formed a sister group to the above-mentioned superclade while DENV-1 formed the most basal clade of the entire tree.

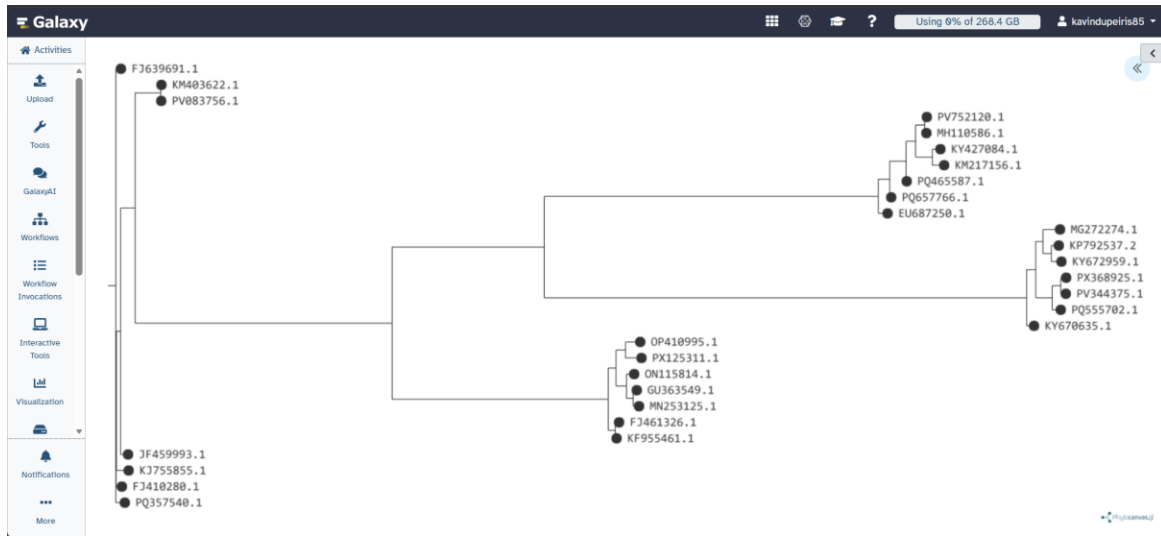


Figure 8. Consensus Maximum Likelihood phylogenetic tree of 28 Asian DENV isolates (1,000 ultrafast bootstrap replicates), color-partitioned by serotype.

3.5 Geographic and Temporal Clustering Within Serotypes

Sub-clustering within each serotype clade usually occurs due to geographical proximity and possible epidemiological connection. For DENV-1, isolate KM403622.1 (Singapore, 2013) and PV083756.1 (Guangzhou, China, 2014) were clustered together with 100% bootstrap support indicating that a possible transmission connection exists between Southeast Asia and southern China. In DENV-2, isolates collected from Sri Lanka (2017) and from China (2017) belonged to the same well-supported clade (100%), while isolates collected from India (2010) and Pakistan (2011) belonged to another well-supported sub-lineage (100%). The two sub-lineages along with isolates collected from Malaysia and Bangladesh formed a larger clade with 96-99% bootstrap support depending on the depth of the bootstrap used. Isolates of DENV-3 split into an East/South Asian lineage where isolate GU363549.1 (China) and isolate MN253125.1 (India) clustered together with 100% bootstrap support and into an Indochinese lineage connecting isolates from Vietnam and Cambodia. For DENV-4, two main subgroups have been observed one connecting isolate from Singapore (2011) and China (2015),

4. Discussion

4.1 Why is sequence alignment required before phylogenetic analysis?

The process of sequence alignment determines the positional homology, where each nucleotide in a certain column in all the sequences comes from the same evolutionary position. This is because the process of insertions and deletions changes the register of the sequence with regard to the others. Therefore, if the comparison was done without alignment, it would mean that non-homologous positions were compared. Through alignment, gap characters are put in the positions of insertions and deletions, which means that each column represents one evolutionary character. In the absence of alignment, both distance and model-based phylogenetic methods will not be able to interpret indels properly, and consequently construct wrong trees.

4.2 What is the purpose of bootstrap analysis?

The bootstrapping analysis technique is one of the resampling techniques that estimate how strongly supported a given branch/clade in the phylogenetic tree is by the underlying data. Resampling is done on the columns of the original alignment and many pseudo-replicate alignments of equal lengths are generated using the same columns, and a tree is constructed from each of these pseudo-replicates. The bootstrap value for any given clade is estimated using the percentage of replicate trees in which this clade is found. The bootstrapping value does not provide the probability of whether the clade is indeed correct but provides the confidence level on the basis of the existing data.

4.3 How does Maximum Likelihood differ from distance-based methods?

Distance-based and Maximum Likelihood (ML) methods are inherently dissimilar approaches to phylogenetic inference. While Neighbor-Joining and other distance-based techniques first reduce the alignment to a genetic distance matrix, which is calculated based on a relatively straightforward substitution model like Jukes-Cantor or Kimura two-parameter is, and then try to build a single tree from the resulting matrix in such a way that the total branch length is minimized, ML algorithms examine the alignment position by position according to a statistical substitution model with specific parameters including base frequencies, transition/transversion ratio, and rate heterogeneity and find those tree structure and branch lengths that give the maximum probability of obtaining the observed data. Therefore, ML is substantially more time-consuming than the distance-based approach, but at the same time, it is also more accurate and reliable in cases of, for example, long-branch attraction and high rate heterogeneity both of which are common among rapidly evolving RNA viruses such as DENV.

4.4 What biological conclusions can be drawn from the generated tree?

This tree offers multiple biological implications of DENV transmission within the Asian region. First, the simultaneous circulation of all four serotypes in individual countries like China, India, and Singapore implies high endemicity of the region along with a heightened threat of severe disease through antibody-dependent enhancement when infected a second time by another serotype. Second, the existence of

several well-established clades reveals multiple cases of successful transnational transmission: for instance, the grouping of the Singapore isolate with the Guangzhou, China isolate within DENV-1 (100% support) is indicative of one such transmission event connecting Southeast Asia and southern China, as well as the tight Sri Lanka-China clade within DENV-2 is indicative of rapid transmission of that lineage internationally. Third, the existence of two South Asian sublineages of DENV-2 (India and Pakistan) shows that transmission within the region is not sporadic and frequent, but localized and ongoing. Overall, this pattern fits the region where dengue virus lineages both circulate endemically in each country and are periodically transmitted between countries.

4.5 Limitations of the analysis

- **Sampling bias:** 28 genomes are a tiny sample of the millions of cases of dengue infections that happen yearly all over Asia, limiting the accuracy in pinpointing the origin of outbreaks and tracing transmission pathways.
- **Metadata:** the lack of comprehensive clinical and travel history metadata available for most isolates restricts the epidemiological implications that could have been concluded based on clade affiliation only.
- **Recombination:** IQ-TREE relies on the assumption of strict bifurcation of tree branches, while RNA viruses such as DENV may experience recombination; in case it does happen, it would affect branch length and topology estimation inaccurately.
- **Assembly issues:** misassembly issues, especially those concerning untranslated regions at 5' and 3' ends of the genome, may produce erroneous results in the alignment and subsequent tree reconstruction.

4.6 Bonus challenge: 100 versus 1,000 bootstrap replicates

For assessing the influence of the number of bootstraps used on the reliability of the estimated phylogeny, the entire procedure was performed with 100 and with 1,000 fast bootstrap replicates, and support values were calculated for a sample of nodes (Table 5).

Table 4. Comparison of bootstrap support values under 100 versus 1,000 replicates for representative nodes.

Clade	100 reps	1000 reps	Shift
DENV-1 / DENV-2 / DENV-3 / DENV-4 monophyly	100%	100%	Equal
DENV-2 + DENV-4 sister grouping	100%	100%	Equal
KM403622.1 (Singapore) + PV083756.1 (China), DENV-1	100%	100%	Equal

Clade	100 reps	1000 reps	Shift
South Asian sub-clade + PQ465587.1 (Malaysia), DENV-2	96%	99%	+3%
GU363549.1 (China) + MN253125.1 (India), DENV-3	98%	100%	+2%
OP410995.1 (Singapore) + PX125311.1 (Indonesia), DENV-3	80%	95%	+15%
PX368925.1 + PV344375.1 cluster, DENV-4	94%	100%	+6%
DENV-1 internal backbone node	41%	64%	+23%
DENV-4 Subgroup A / B backbone split	64%	75%	+11%

Nodes distinguishing the four serotypes were well-resolved with 100% support regardless of replicate depth, due to the large genetic divergence among serotypes. The saturation of these nodes means they cannot be improved with increased replicate depth. On the other hand, some nodes within serotypes experienced improvement of support values by increasing replicate number, e.g., the clade containing a Singaporean and an Indonesian isolate of DENV-3 increased its value from 80% to 95%, and the backbone node of DENV-1 increased from 41% to 64%. The explanation for this phenomenon is that the standard bootstrapping method based on few replicates may be more vulnerable to sampling error in the case of less reliable nodes, while the bootstrap analysis with greater replicates provides more robust results which better reflect the true support value.

5. Conclusion

In this study, Maximum Likelihood phylogeny of 28 complete genomes of dengue virus from human blood collected across Asia was generated and interpreted using a highly replicable Galaxy workflow consisting of NCBI Virus sequence data collection, MAFFT alignment, and IQ-TREE tree construction using the TIM2+F+I+R3 model selected by ModelFinder. In the results, all the four DENV serotypes are distinct, well-supported monophyletic clades; serotypes DENV-2 and DENV-4 have been identified as sister serotypes, and there are several highly supported transnational sub-clades in addition to the South Asian sub-lineages. When analyzing the bootstrap analyses performed using 100 and 1,000 replicates, it becomes clear that while deep serotype-level nodes remain robust regardless of the number of replicates used, weakly supported nodes gained significantly when using 1,000 replicates; 1,000 replicates can thus be considered as the optimal number for viral phylogenies. In the broad sense, this study demonstrates how easily accessible, cloud-based pipelines can be used to perform the continuous monitoring of dengue virus evolution and transmission as well as serotype co-circulation in Asia, which can provide important information about the severity of dengue virus infections.

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